

HANHUI WANG

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EDUCATION

Ph.D. in Computer Science

Khoury College of Computer Sciences, Northeastern University (NEU)

Sept.2025 - Present

Advisor: **Prof. Huaizu Jiang**

M.S. in Computer Science

Viterbi School of Engineering, University of Southern California (USC)

Aug.2023 - May 2025

GPA 4.0/4.0

B.Eng. in Computer Science and Technology

Sept.2019 - June 2023

School of Computer Science & Technology (SCST)

Huazhong University of Science and Technology (HUST)

GPA 3.99/4.0 Rank 2/363

★ Selected by SCST for a special class for the most promising students

RESEARCH EXPERIENCE

Research Assistant

June 2024 - Dec.2024

TACO Group, Texas A&M University

Advisor - **Prof. Zhengzhong Tu**

- Presented a novel perspective for protecting personal images from malicious editing, focusing on making biometric features unrecongnizable post editing.
- Developed a new algorithm, **FACELOCK**, that incorporates facial recognition models and feature embedding penalties to effectively protect against diffusion-based image editing.
- Conducted a crucial analysis of the quantitative evaluation metrics commonly used in image editing tasks, exposing their vulnerabilities and highlighting the potential for manipulation to achieve deceptive results.

Research Assistant

May 2024 - Aug.2025

Visual Intelligence Lab, Northeastern University

Advisor - **Prof. Huaizu Jiang**

- Introduced **SNAP**, the first unified segmentation model capable of handling diverse point cloud domains (indoor, outdoor, aerial) while supporting versatile prompt types such as spatial points (clicks) and text for flexible object segmentation.
- Introduced **Struct2D**, a perception-guided prompting framework that unlocks spatial reasoning in Large Multimodal Models (LMMs) using structured 2D representations.

Research Assistant

Sept.2022 - Mar.2023

Embedded and Pervasive Computing Lab, HUST

Advisor - **Prof. Xianzhi Li**

- Modified a Few-shot learning framework for 3D Instance Segmentation (3DIS) to address the problem of the expensive costs of collecting a sufficient amount of annotated point clouds.
- Utilized a Transformer Decoder to generate differentiated kernels to perform instance-wise dynamic convolution.
- Implemented the model using **Python** and **PyTorch** and improved the mean Average Precision results (mAP) by **3.2** percent on the ScanNet V2 dataset.

PUBLICATION & PREPRINT

1. Aniket Gupta*, **Hanhui Wang***, Charles Saunders, Aruni Roy Chowdhury, Hanumant Singh, Huaizu Jiang. **SNAP: Towards Segmenting Anything in Any Point Cloud**. In *Proceedings of the 13th International Conference on 3D Vision (3DV)*, 2026, [Oral](#).

2. Fangrui Zhu*, **Hanhui Wang***, Yiming Xie, Jing Gu, Tianye Ding, Jianwei Yang, and Huaizu Jiang. **Struct2D: A Perception-Guided Framework for Spatial Reasoning in Large Multimodal Models**. In *Proceedings of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
3. **Hanhui Wang***, Yihua Zhang*, Ruizheng Bai, Yue Zhao, Sijia Liu, and Zhengzhong Tu. **Edit Away and My Face Will Not Stay: Personal Biometric Defense against Malicious Generative Editing**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
4. **Hanhui Wang**, Huaize Ye, Yi Xia, Xueyan Zhang. **Leveraging SAM for Single-Source Domain Generalization in Medical Image Segmentation**. In *arXiv 2401.02076*, 2024.

COMMUNITY SERVICES

- Conference Reviewer: CVPR, CVPR Workshop, 3DV, ECCV

TALKS

- [SNAP](#): Towards Segmenting Anything in Any Point Cloud. Feb. 2026

TEACHING EXPERIENCE

- Teaching assistant of CS7150, Deep Learning, Northeastern University. Spring 2026

PROJECTS

Stock Management Android App Apr.2024 - May 2024

- Developed a stock management Android Application using Android Studio.
- Refactored the **Node.js** backend to make it suitable for an Android app.
- Utilized **Volley**, **Picasso** to handle multiple HTTP requests.

Stock Management Website Mar.2024 - Apr.2024

- Developed and deployed a stock management [website](#) on the Google Cloud Platform (GCP).
- Designed and implemented the frontend service using HTML5, Bootstrap, and **Angular** frameworks. Ensured a responsive and user-friendly interface to enhance user experience and engagement.
- Engineered the backend service using **Node.js**, ensuring robustness, scalability, and high performance. Implemented server-side logic, database integration, and API handling to support frontend functionalities.
- Managed and maintained a cloud-based **MongoDB** database to securely store and manage data.

Single-Source Domain Generalization project Oct.2023 - Dec.2023

- Led a group of 4 to work on a research project focused on leveraging the Segment Anything Model (SAM) for Single-Source Domain Generalization in the context of Medical Image Segmentation.
- Proposed a dual-stage fine-tuning paradigm for SAM to address domain generalization tasks.
- Designed an efficient mask-filtering module to generate refined bounding boxes for SAM.
- Our approach achieved state-of-the-art results on the Prostate dataset, **8** percent higher than the former best results.
- Organized the creation of the poster and drafting of the [paper](#).

Jigsaw Puzzle project Nov.2021 - Dec. 2021

- Led a group of 4 to develop a jigsaw puzzle game-playing website using **JavaScript** and the **Paper.js** graphics framework, and maintained the project repository at [Gitee](#). ([Video Demo](#))
- Designed a novel magnetic mode for fun-seeking users.

- Deployed the website on a Cloud Server and developed the back-end services to deal with the images uploaded by different players.
- Adapted the website and game-playing operations to different PC and mobile devices.

SELECTED AWARDS & HONORS

China National Scholarship (the highest national wide scholarship for undergraduate students in China)	2020
Outstanding Undergraduates in Term of Academic Performance (the greatest honor for undergraduates in HUST)	2020
Merit Student of HUST	2020, 2021, 2022
Outstanding Graduates of HUST	2023